

SHAMBHURAJ MANE

LabVIEW Application Engineer

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PROFESSIONAL SUMMARY

LabVIEW Software Developer with 3 years of experience developing industrial test and automation systems. Experienced in the design, development, installation, and commissioning of software for End-of-Line (EOL) test systems and data acquisition applications. Skilled in integrating LabVIEW with PLCs, servo drives, sensors, custom embedded hardware, and SQL databases, enabling automated testing, motion control, and data management solutions. Interested in opportunities in EOL Testing, Automated Test Equipment (ATE), Hardware-in-the-Loop (HIL), and industrial test automation.

CORE SKILLS

- NI LabVIEW
- NI MAX
- NI OPC Servers
- SQL
- NI cDAQ
- NI cRIO
- NI USB DAQ
- NI PXI Chassis
- TCP/IP
- NI Ethernet/IP
- RS-232/RS-485
- OPC/OPC UA
- PLC Integration (Siemens, Allen Bradley)
- Keyence Scanners (SC-X Series)
- PID Controllers (Autonics)

PROFESSIONAL EXPERIENCE

LabVIEW Application Engineer · *Theta Measurement and Control Solutions Pvt. Ltd.*

Aug 2023 – Present

Pune, Maharashtra

- LabVIEW Application Development and Software Debugging
- Interfacing with NI Hardware like cDAQ, cRIO, USB DAQ, PXI Chassis for Data Acquisition and Control. Also Interfacing with third party Hardware such as PLC, PID using communication protocols like Modbus, TCP/IP, Ethernet-IP, USB, RS-232, RS-485, OPC/OPC UA etc.
- Onsite Installation and Commissioning.

Key Projects: -

1. Gear Roll Tester: Industrial Gear Quality Inspection System

- Developed a LabVIEW-based industrial gear inspection system integrating custom embedded hardware, Baker LVDT probes, encoders, and Mitsubishi servo drives.
- Implemented automated motion control using Modbus communication and Mitsubishi MR-JE-40A servo drives.
- Developed defect detection and gear analysis algorithms for burrs, tooth irregularities, jerky gear behavior, TCE, TTCE, PCD, and OBD measurements.

2. Power Windows Regulator Testing: An EOL (End-of-the-Line) Test System for Quality Inspection of Power Windows Regulator

- Integrated Allen-Bradley PLCs, Keyence SR-X barcode scanners, and PWR motors with LabVIEW using Ethernet/IP, TCP/IP, and LIN protocols.
- Developed EOL test software features including recipe management, diagnostics, communication configuration, user authentication, and access control.
- Developed LIN communication modules and automated traceability solutions for production-line testing.

3. 150-Channel Data Acquisition System for Hindustan Aeronautics Ltd:

- Developed a LabVIEW-based 150-channel data acquisition system with channel configuration, calibration, and user-configurable acquisition settings.
- Implemented support for variable sampling rates, acquisition frequencies, and load sensor excitation control.
- Developed a TCP/IP client-server architecture using the STM Library for remote monitoring and control.

4. Shim Selection Software for Engine Assembly Line:

- a. Developed the software That can select Shims for the engine, the software has features such as calibration and access control
- b. Developed a Software Module That communicates with siemens S7-1200 PLC using OPC Protocol

5. Shell Bearing Tester for Engine Assembly Line:

- a. Developed the PLC Communication Module that performs communicates with and Performs handshaking of LabVIEW with Simatic S7-1200 PLC using S7-Protocol by using S7-Net Comm Pallet of LabVIEW
- b. Developed a Test sequence in LabVIEW that conducts the test cycle and stores the data in the end of test cycle

6. EOL(End-of-the-Line) Noise Testing for car Axles

- a. Contributed to the Development as well as Installation and Commissioning of the software
- b. Developed a Software Module That Communicates with Siemens S7-1500 PLC via OPC UA Protocol
- c. Developed a Test Sequence in LabVIEW That conducts the test cycle and displays the results on the screen
- d. Developed Software Modules that Communicate with inisys Shift PC and ipattern NVH PC using RS-232 Protocol

Trainee Application Engineer · *Theta Measurement and Control Solutions Pvt. Ltd.*

Dec 2022 – June 2023

Pune, Maharashtra

- LabVIEW Application development and Software debugging
- NI Hardware interfacing and development of data acquisition applications

EDUCATION

BE in Instrumentation and Control · *AISSMS Institute of Information Technology, Pune*

2019 - 2023